Hunsoo Song, Ph.D.

Assistant Professor, School of Civil Engineering
Chungbuk National University (CBNU)
1 Chungdae-ro, Seowon-gu, Cheongju, Chungbuk 28644, Republic of Korea
Email: hunsoo.song@cbnu.ac.kr | Phone: +82-10-6408-5944
Lab Website (GSEL): https://gsel.xyz

RESEARCH INTERESTS

Geospatial Data Science, Urban Sustainability, Climate Change, etc.
Current focus: Analysis of infrastructure and environmental inequality in cities.

APPOINTMENTS

Assistant Professor, Chungbuk National University — Mar 2025 - Present
Director, Geospatial Science & Engineering Lab (GSEL), School of Civil Engineering

Postdoctoral Researcher, Yale University — Sept 2024 - Feb 2025
Hixon Center for Urban Sustainability & Seto Lab, Director: Dr. Karen Seto

Postdoctoral Researcher, Purdue University — June 2024 - Aug 2024
Geospatial Data Science Lab, Director: Dr. Jinha Jung

Graduate Researcher, Oak Ridge National Laboratory — May 2023 - Jul 2023
GeoAI group, Geospatial Science and Human Security Division

Graduate Research Assistant, Purdue University — Aug 2020 - May 2024
Geospatial Data Science Lab, Director: Dr. Jinha Jung

Graduate Research Assistant, Seoul National University — Mar 2018 - Feb 2020
SPINS-RS Lab, Director: Dr. Yongil Kim

EDUCATION

Purdue University, West Lafayette, IN — Aug 2020 - May 2024
Ph.D. in Geomatics, Civil Engineering
Concentration in Computational Engineering
Dissertation: Transparent and Scalable Knowledge-based Geospatial Mapping Systems for Trustworthy Urban Studies

Seoul National University, Seoul, Korea — Mar 2018 - Feb 2020
M.S. in Civil and Environmental Engineering

Seoul National University, Seoul, Korea — Mar 2012 - Feb 2018
B.S. in Civil and Environmental Engineering

SCHOLARSHIPS AWARDS

Podium Presentation Award (2nd place), 28th Environmental Engineering & Science Symposium, Champaign, IL, 2023.

GISCUP Winner (1st place), 30th ACM SIGSPATIAL, Seattle, WA, 2022.

Roland S. Corning II Fellowship, Purdue University, 2022.

Frederick N. Andrews Fellowship, Full Tuition & Stipend Coverage, Purdue University, 2020-2024.

Student Competition using Meteorological Satellites (2nd place), Korea Meteorological Administration, 2019.

Best Student Paper Award, International Symposium on Remote Sensing, 2019.

Merit-based Scholarship, Seoul National University, 2018.

Brain Korea 21 Plus Scholarship, National Research Foundation of Korea, 2018.

Certificate of Commendation, Korean Society of Survey, Geodesy, Photogrammetry, and Cartography, 2018.

Best Bachelor's Thesis Award (3rd place), Department of Civil and Environmental Engineering, Seoul National University, 2017.

National Scholarship for Science and Engineering, Full Tuition Coverage, Korea Student Aid Foundation, 2014-2017.

Merit-based Scholarship, Seoul National University, 2012-2013.

REFEREED JOURNAL ARTICLES

(*Corresponding author)

[P12] Yeonjae Kim, **Hunsoo Song***, Songlin Fei, Jinha Jung (2026), "Transforming leaf-off LiDAR to leaf-on canopy height models using deep learning" *GIScience & Remote Sensing, 63(1).*

[P11] Jungoung Kim, **Hunsoo Song*** (2026), "How reliable are geospatial foundation model embeddings for cross-city urban mapping?" *IEEE Geoscience and Remote Sensing Letters, 23, 5003105.*

[P10] **Hunsoo Song**, Jinha Jung (2025), "Unsupervised surface water mapping with airborne LiDAR data by leveraging physical properties of water" *GIScience & Remote Sensing, 62(1), 2437252.*

[P9] Arnav Goel, **Hunsoo Song**, Jinha Jung (2025), "Integrating sparse LiDAR and multi-sensor time series imagery from spaceborne platforms for deriving localized canopy height model" *IEEE Transactions on Geoscience and Remote Sensing.*

[P8] Dennis Choi, Lindsay E Darling, Jaeyoung Ha, Jinyuan Shao, **Hunsoo Song**, Songlin Fei, Brady Hardiman (2025), Understanding the effects of spatial scaling on the relationship between urban structure and biodiversity." *International Journal of Applied Earth Observation and Geoinformation, 138, 104441.*

[P7] **Hunsoo Song**, Anamika Shreevastava, Gaia Cervini, Jinha Jung (2024), "Reshaping landscape factorization through 3D landscape clustering for urban temperature studies." *Sustainable Cities and Society, 115, 105809.*

[P6] Hansae Kim, **Hunsoo Song**, Jinha Jung (2024), "Cadastral-to-Agricultural: A Study on the Feasibility of Using Cadastral Parcels for Agricultural Land Parcel Delineation." *Remote Sensing.*

[P5] **Hunsoo Song**, Jinha Jung (2023), "An unsupervised, open-source workflow for 2D and 3D building mapping from airborne LiDAR data" *In IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing.*

[P4] **Hunsoo Song**, Jinha Jung (2023), "An object-based ground filtering of airborne LiDAR data for large-area DTM generation." *Remote Sensing*, 15, 4105.

[P3] **Hunsoo Song**, Lexie Yang, Jinha Jung (2022), "Self-filtered learning for semantic segmentation of buildings in remote sensing imagery with noisy labels." *In IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16, 1113-1129.

[P2] Minho Kim, **Hunsoo Song**, Yongil Kim (2020), "Direct short-term forecast of photovoltaic power through a comparative study between COMS and Himawari-8 meteorological satellite images in a deep neural network." *Remote Sensing*, 12(15), 2357.

[P1] **Hunsoo Song**, Yonghyun Kim, Yongil Kim (2019), "A patch-based light convolutional neural network for land-cover mapping using Landsat-8 images." *Remote Sensing*, 11, 114.

REFEREED CONFERENCE PROCEEDINGS

[C10] **Hunsoo Song**, Lexie Yang (2024), "Efficient extraction of building elevation attributes for flood risk management using airborne LiDAR data." *In 2024 IEEE International Geoscience and Remote Sensing Symposium. IEEE.*

[C9] Chenying Liu, **Hunsoo Song**, Anamika Shreevastava, Conrad Albrecht (2024), "AUTOLCZ: Towards automatized local climate zone mapping from rule-based remote sensing." *In 2024 IEEE International Geoscience and Remote Sensing Symposium. IEEE.*

[C8] **Hunsoo Song**, Joshua Carpenter, Jon E. Froehlich, Jinha Jung (2023), "Accessible Area Mapper for inclusive and sustainable urban mobility: a preliminary investigation of airborne point clouds for pathway analysis." *In Proceedings of the 1st ACM SIGSPATIAL International Workshop on Sustainable Mobility*, Hamburg, Germany, Nov 13-16.

[C7] **Hunsoo Song**, Gaia Cervini, Jinha Jung (2023), "Assessment of local climate zone products via simplified classification rule with 3D building maps" *In 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE.* Pasadena, CA, Jul 16-21.

[C6] **Hunsoo Song**, Jinha Jung (2022), "Challenges in building extraction from airborne LiDAR data: ground-truth, building boundaries, and evaluation metrics" *In the 30th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.*, Seattle, WA, Nov 1-4.

[C5] **Hunsoo Song**, Gwangjoong Kim, Minho Kim, Yongil Kim (2019), "Short-term forecasting of photovoltaic power integrating multi-temporal meteorological satellite imagery in deep neural network." In *2019 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC)*. IEEE. Macau, China, Dec 1-4.

[C4] Gwangjoong Kim, **Hunsoo Song**, Minho Kim, Yongil Kim (2019), "Multimodal merging of satellite imagery with meteorological and power plant data in deep convolutional neural network for short-term solar energy prediction." in *Asian Conference on Remote Sensing 2019*, Daejeon, Korea, Oct 14-18.

[C3] **Hunsoo Song**, Anjin Chang, Junho Yeom, Jinha Jung, Yongil Kim (2019), "Domain adaptation for 2D/3D change detection in VHR imagery via calibration of convolutional neural network under prior probability shift." in *Asian Conference on Remote Sensing 2019*, Daejeon, Korea, Oct 14-18.

[C2] **Hunsoo Song**, Yongil Kim (2019), "Improving land-cover classification accuracy with a patch-based convolutional neural network: data augmentation and purposive sampling." In *2019 Joint Urban Remote Sensing Event (JURSE)*. IEEE. Vannes, France, May 22-24.

[C1] **Hunsoo Song**, Yongil Kim (2019), "A patch-based supervised approach for change detection in high resolution multispectral images." in *International Symposium on Remote Sensing 2019*, Taipei, Taiwan, Apr 17-19.

Research Grant Acquisition

Citizen-Centric Social-Safety Smart City (Education & Research Group) — (Ministry of Education & National Research Foundation of Korea, Brain Korea 21 (BK21) Four, Regional University Consortium Pilot Program) Sep 2026 - Aug 2027

- Serve as *Participating Faculty* in a four-university regional consortium led by Chungbuk National University (with Hanbat National University, Daejeon University, and Cheongju University)

Developing a Korean Urban Embedding through Foundation Model Localization — (National Research Foundation of Korea, Young Researcher Program (Type B), No. RS-2026-25499133, KRW 700,320,000) Mar 2026 - Feb 2031

- Serve as *Principal Investigator*

Developing an Airborne LiDAR-Based Urban Walkability Assessment Model to Address Infrastructure Accessibility Inequalities — (National Research Foundation of Korea, Outstanding Young Researcher Program (Seed Research), No. RS-2025-24803224, KRW 92,122,000) Sep 2025 - Aug 2026

- Served as *Principal Investigator*

Automated Detection of Ground Control Points Considering Positional Characteristics of Satellite Imagery — (Korea Aerospace Research Institute) May 2025 - Dec 2025

- Served as *Principal Investigator*

A Scalable and Sustainable Framework for a Geospatial Digital Twin — (National Geospatial-Intelligence Agency, $378,215) Jul 2023 - Jun 2025

- Played a pivotal role in acquiring the proposal — *led proposal writing*
- Serve as the primary researcher for this project

Punjab Urban Land Systems Enhancement Project — (Food and Agriculture Organization of the United Nations, Total $40,000) Mar - Jun 2022, Mar - Jun 2023

- Contributed to proposal writing and methodology development

Photovoltaic Power Estimation using Meteorological Satellite Imagery — (SK Telecom—Korea's largest mobile operator, ∼$59,000) Jul 2019 - Dec 2019

- Served as *project manager*, leading the proposal and final report writings, and directing the entirety of the research efforts
- Achievements include: 1 patent, 1 journal article, 2 conference proceedings

Patents

Hunsoo Song, Yongil Kim, Minho Kim, Gwangjoong Kim, "Method and Apparatus for Short-term Photovoltaic Power Prediction Based on Convolutional Neural Network", South Korea Patent Application No. 10-2021-0008489, filed Jan 21, 2021.

Hunsoo Song, Yongil Kim, "Apparatus and Method for Generating Land Cover Map", South Korea Patent Application No. 10-2019-0095402, filed Aug 6, 2019.

Invited Talks

"Designing Smart Cities with 3D Urban Digital Twins: Accessibility and Urban Heat Modeling", AI and Smart Cities Seminar, Division of Smart Cities, Korea University, Sejong, Republic of Korea Jun 2026

"Satellite Imagery and 3D Geospatial Analytics for Renewable Energy and Carbon Monitoring", Graduate Seminar, Energy Process Innovation Convergence Major, Inha University, Online Apr 2026

"Development and Application of Knowledge-Based GeoAI Mapping Sys-

tems for Trustworthy Geospatial Digital Twins", Institute of Construction Technology (IOCT), Chungbuk National University, Seoul, South Korea Apr 2025

"Toward Trustworthy GeoAI Systems for Geospatial Digital Twin", BK21 InfraSPHERE Seminar, Seoul National University, Seoul, South Korea Apr 2025

"Trustworthy Geospatial Digital Twins and Simulations for Urban Heat Studies", Mui Ho Center for Cities, Cornell University, Ithaca, NY May 2024

"Physical Property-Driven 3D Terrain and Surface Water Mapping", The U. S. Geological Survey — CEGIS Annual Meeting, Rolla, MO Aug 2023

"Airborne LiDAR for Digital Twin: Advancing Scalable Urban Resilience Research", SI Analytics, Online seminar Aug 2023

PROFESSIONAL AFFILIATIONS

Member, IEEE GRSS Image Analysis and Data Fusion Technical Committee (IADF TC) 2026–Present

Member, International Association for Urban Climate (IAUC) 2026–Present

Member, Future Earth Urban Knowledge–Action Network 2026–Present

TEACHING & MENTORING SERVICES

Instructor of Record, Chungbuk National University, 2025 - Present

- *Undergraduate:* Geo-spatial Information Analysis (Spring 2025, Spring 2026); Photogrammetry — Capstone Design (Fall 2025, Fall 2026)
- *Graduate:* Digital Photogrammetry (Spring 2025); Advanced Photogrammetry (Fall 2025); Data Fusion and Artificial Neural Networks (Spring 2026); Spatial Data and Deep Learning (Fall 2026)

Academic Mentoring, Various periods, 2021 - Present

- Provided academic and research guidance to undergraduate and graduate students across diverse disciplines
- Advisee Jimin Hwang (M.S. student) received the *Best Student Paper Award*, Student Paper Competition, 2026 KSRS Fall Conference, Busan, Oct 2026

Teaching Assistant, CE 203: Principles and Practice of Geomatics, Purdue University, Fall 2021 & Fall 2023

- Served as Lead Teaching Assistant for a class of more than 140 undergraduate students
- Delivered lectures and laboratory sessions and contributed to course-material development

Volunteer Judge, Undergraduate Research Conference, Purdue University, Spring & Summer 2022

- Evaluated undergraduate research projects and provided academic feedback

Young Engineers Honor Society, National Academy of Engineering of Korea, Nov 2016 - Aug 2020.

- Engineers selected by university heads from various majors
- Mentored diverse student bodies, volunteered to teach basic engineering courses, and engaged in academic exchanges and social contribution initiatives

Military Service, South Korea, Sept 2014 - Jun 2016.

REFERENCES

Jinha Jung, Purdue University
Associate Professor, Civil Engineering
jinha@purdue.edu

Karen Seto, Yale University
Frederick C. Hixon Professor of Geography and Urbanization Science
karen.seto@yale.edu

Melba M. Crawford, Purdue University
Nancy Uridil and Frank Bossu Distinguished Professor, Civil Engineering
mcrawford@purdue.edu

Hsiuhan Lexie Yang, Ph.D., Oak Ridge National Laboratory
Research Scientist, GeoAI Group
yangh@ornl.gov

Songlin Fei, Purdue University
Professor, Forestry and Natural Resources
sfei@purdue.edu